

Harmonizing Abilities: Collaborative Digital Musical Instruments for Students with Complex Disabilities Using the Some for Many Strategy

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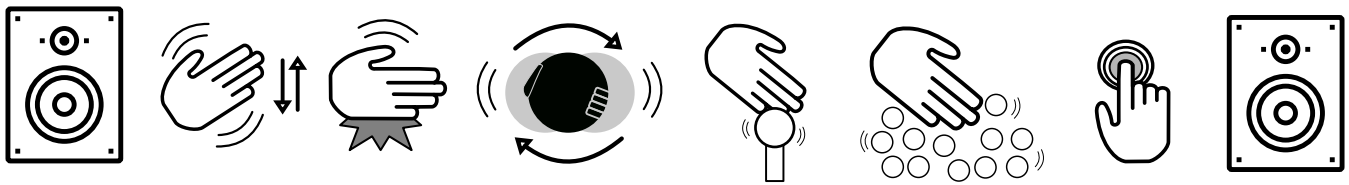


Figure 1: Six typical interactions identified for the design of our instruments

ABSTRACT

In our case study, we propose a ‘Some for Many’ design approach for digital musical instruments (DMIs) in special educational needs (SEN) schools that addresses the challenges of designing for and with students with complex disabilities and non-verbal communication. We outline the development and evaluation of several DMIs based on typical everyday interactions. Our findings suggest that the approach is likely to promote active music-making and inclusion and potentially contributes to a change in peers’ perceptions of students with complex disabilities in a positive way.

CCS CONCEPTS

• **Applied computing** → **Sound and music computing**; • **Human-centered computing** → **Accessibility technologies**; **Participatory design**.

KEYWORDS

Accessible Digital Musical Instruments, Music Technology, Tangible Interaction, Open Source, Special Education, Pure Data, Bela

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1 INTRODUCTION

Since the ratification of the UN Convention on the Rights of Persons with Disabilities [21], active music-making must be considered as a basic human right [11, 12]. In addition to making music as an end in itself, active music-making can also benefit non-musical competencies, particularly relevant in the context of special educational needs (SEN) schools. For example, making music can fundamentally have a positive effect on well-being and social-emotional development [6, 19, 23, 31]. In addition, music making can, for example, improve communication and participation in groups [2], and likewise positively influence attitudes of children toward the phenomenon of disability, where an essential prerequisite seems to be enjoyable musical activities with musicians with disabilities perceived as competent [20]. A prerequisite for active music-making is the availability of appropriately accessible musical instruments. The problem is that conventional acoustic musical instruments may be experienced as inaccessible, especially by students with complex disabilities (for an explanation of this term, see section 3.1) [22]. Digital musical instruments (DMIs) may be an option to facilitate access to active music-making [4, 10, 11, 25], but have been little used in the SEN context [11].

For this reason, we spent a year developing DMIs in a German SEN school as part of a participatory design project, for example, to provide access to loop-based music making [15]. As we were working with a very heterogeneous group of children, these instruments were not fully accessible to everyone. This was mainly due to the abstract interaction metaphors and indirect interactions. Especially for the group of children with complex disabilities, a different design approach was necessary. Therefore, we developed the idea of ‘Some for Many’ (detailed in section 3). In this paper,

we present this design approach and discuss the instruments we developed during the project, with a particular focus on students with complex disabilities. We then present our evaluation of these instruments, drawing on insights from qualitative interviews with the music teachers, our own observations, and questionnaires from the children's personal assistants.

2 RELATED WORK

There is a research gap in the literature regarding the use of DMIs in the SEN school context [8, 10, 11]. For Germany, we found that DMIs were generally rarely used, but teachers were optimistic about their potential [11]. Teachers cited *ease of use* and *applicability in group settings* as essential requirements for DMI use. To date, little work has been done to create accessible DMIs for collaborative music-making in group settings [12, 24]. A lack of technical and general knowledge have been repeatedly identified as the main barriers to DMI use, as well as the high cost of commercially available DMIs [3, 5, 11]. An overview of available accessible DMIs was provided by Frid [12] and of DMIs for “clients with complex needs” by Ward et al. [26]. In addition, design criteria for the SEN context have been presented in the literature, emphasizing the need for the inclusion of the target group itself in the development process [11, 14, 18, 27].

Both Ward [28] and Lindetorp et al. [17] followed a design approach similar to the one we propose. In her dissertation, Asha Ward used several iterations to develop a set of controllers that connect to the MAX software. Except for her *TouchBox* instrument, which works standalone with a built-in speaker, the instruments need to be connected to a computer. Lindetorp et al. also developed several instruments for collaborative music-making. The instruments were based on earlier developments presented in Svahn et al. [24] and “tried to match the sounds with the actions as closely as possible to acoustic instruments” [17, p. 5]. Furthermore, the instruments have to be connected to a “master instrument” that needs assistants for synchronization. They concluded that “music composed for this setup should allow for unsynchronized, unplanned, and irregular actions” [p. 13].

During our project, the school's speech and language therapist told us that in his experience, even having to set up a simple Bluetooth connection led to the non-use of devices at school. This was consistent with findings of technical barriers in the literature [5, 11]. Thus, we chose a sound design that provided ambient-oriented music because the children in our group enjoyed the calm and relaxing vibe of this music genre, and it allowed for unsynchronized use of the instruments, eliminating the need for a central control unit and reducing possible technical barriers for teachers. Since we found in previous studies with music teachers that the reference to acoustic musical instruments could contribute to a ‘deficit-oriented’ view of DMIs [11, 14], which has been criticized in the literature [4], we refrained from trying to adapt the instruments to acoustic models.

3 DESIGN APPROACH: SOME FOR MANY

In a qualitative survey of music teachers in SEN schools in Germany [14], one music teacher stated that ideally one would develop individual bespoke instruments for each student. However, he felt that this was not practical, because of the limited personal and financial resources available. Therefore, he felt that there should

be a variety of simple instruments with different affordances from which the most appropriate one could be chosen with the students individually. This preference was also confirmed by the majority of music teachers in our subsequent large-scale nationwide survey [11]. Based on this idea, we developed the ‘Some for Many’ design approach, which we outline below.

3.1 Target population

The term ‘complex disability’ lacks consistency in its usage. For this study, we adopt Fornefeld's [9] description of complex disability as a group of individuals with heterogeneous abilities and personalities who are homogeneously at risk of exclusion. They rely heavily on caregivers, but due to their limited verbal communication, caregivers may experience difficulties in fully understanding their needs. Hence, these individuals frequently encounter rejection. In music education, active participation in music-making is usually limited. Naturally, such labeling provides little insight into individual abilities. The challenge faced by teachers, therapists, and the instrument designer in the design for the children in our group was to determine the intentionality of the children's actions. Moreover, the children communicated mainly non-verbally, making the medium of language unsuitable for instrument evaluation.

Therefore, conventional participatory design methods like verbal feedback on prototype ideas, discussions, observing children using existing instruments, using low-tech materials, and joint idea elaboration [7] were infeasible. Further, common methods in participatory design for the user group of children with disabilities, like using materials from augmentative and alternative communication (AAC) and the inclusion of proxies [13], were relatively ineffective in involving the students of our group in the design process. Not only because their caregivers experienced difficulties in understanding the children's needs but also because we wanted to focus on the children as much as possible.

3.2 Design Approach

The Some for Many concept aims to bridge the gap between the challenging extremes of ‘design for all’ and ‘design for one’ as illustrated in figure 2. Recognizing the challenge of achieving universal design “which suits door handles, knives, and building entrances but may be more difficult to employ with interactive computer systems” [30, p. 9:5], and the limitations of available resources in SEN schools, this approach focuses on developing a range of bespoke instruments with the idea of meeting the diverse needs of a broader user base. As universal design for learning, an educational paradigm that aims to meet the diverse needs of learners by providing a variety of options for representation, participation and communication, Some for Many aims to develop different instruments that address different individual abilities and preferences. Of course, this approach “cannot account for all of the possible permutations of barriers that an individual might encounter in any given teaching and learning context” [1, pp. 659-660] and there may still be a need to design bespoke instruments for individual students. However, a set of instruments designed with this approach can benefit a wide range of users, balancing complexity, cost, and accessibility in the design of DMIs for the SEN school environment.

Some for Many is not a novel approach in this regard (the basic idea can be found in principles like “Solve for One, Extend to Many.” [16, p. 13]), but rather integrates ideas from both ‘One for All’ and ‘One for One’ approaches with the constraints of the SEN school environment. We draw on ability-based design principles [30] such as focusing on ability rather than *dis*-ability and responding “to poor performance by changing systems, not users, leaving users as they are” [30, 9:8]. The prototypical procedure is to first identify typical everyday interactions of the children and to design instruments that afford musicking based on these interactions (in several iterations if necessary) as shown in figure 3. The objective is to create a collection of independently functioning DMIs that provide direct feedback and can ultimately be used in group music-making.

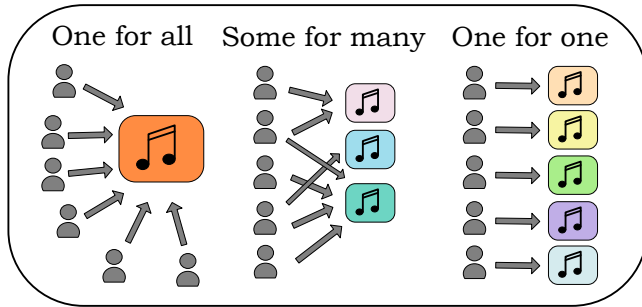


Figure 2: Some for many design approach

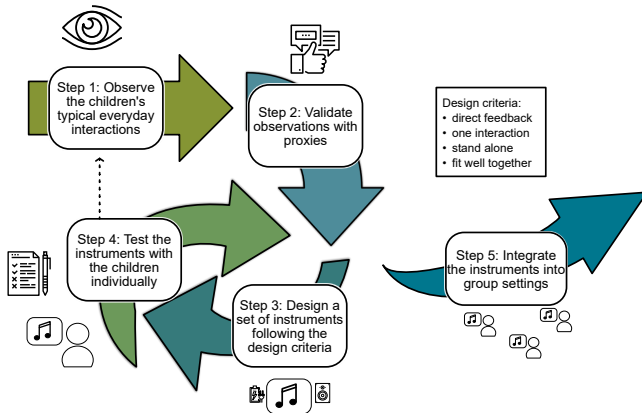


Figure 3: Some for many design steps

4 CASE STUDY

4.1 Schools Project and Participants

The project was conducted for one year in a SEN school with a focus on physical development as a weekly music workshop. Eleven children of varying abilities participated in this workshop, along with their personal assistants, two music teachers, and the first author of this paper. The instruments discussed in this paper were primarily designed for six of these students, who will be briefly introduced later.

All children participated voluntarily after being informed about the project by their teachers and personal assistants and given a choice of different workshops. However, because the focused group of students communicates predominantly nonverbally, obtaining direct consent was challenging. Therefore, we closely monitored nonverbal cues in each session and activity and promptly stopped any activity if there were signs of refusal. Consent was also obtained from parents, teachers, and the school administration. To avoid any bias and to approach the children from an ability-based perspective, we did not inquire about diagnoses at the beginning of the project. At the end of the project, we collected this information using a short questionnaire, which was completed by the respective class teachers. The evaluation of the DMIs was done through observations by the first author (the implications of the observations were discussed with a music teacher after each session), qualitative interviews with the music teachers, and short questionnaires at the end of the project completed by the children’s personal assistants.

4.2 Brief description of children

Emma (age unknown)¹ showed great joy in musical activities in the group. She had a preference for ‘soft’ sounds and calm and relaxing music. She felt relaxed when her personal assistant maintained physical contact with her and generally found comfort in the presence of others. No information was provided about her diagnoses and special educational needs. She communicated nonverbally but, according to the school speech therapist, understood “quite a lot.” Because she had limited physical mobility, she struggled with directed movements, and because she often looked up, it was difficult for her to see what she was doing with her hands.

Liam was 18 years old and had a preference for making ‘loud noises’, which he indicated by constantly shaking a rattling, ball-shaped object with his hand. He showed positive reactions, such as laughter, when interacting with others. The school speech therapist mentioned that he was not sure if Liam had an understanding of object permanence. He also felt that Liam may not have an understanding of cause and effect relationships. Liam was diagnosed with a chromosomal deletion, cerebral palsy, and epilepsy. He needed care in all aspects of daily living.

Leo (age unknown) also had a strong affinity for music, especially jazz, which his personal assistant related to his parents’ preference for this style of music. He enjoyed being in the spotlight but could become restless if the attention was taken away from him. During musical activities, he particularly enjoyed using his hand or leg to hit objects. Leo required assistance and care in all aspects of daily living and required constant attention from a personal assistant due to episodes of breath-interruption. He was diagnosed with brain damage, severe coordination problems, and profound cerebral visual impairment.

Mia was 13 years old and also showed joy participating in group music activities. She further enjoyed dancing to music in her wheelchair. During group singing, she could join in with some of the lyrics. In this regard, the school speech therapist noted that occasionally she would speak in full sentences, although it was unclear if there was a specific purpose behind her words. As she often referred to

¹All names are pseudonyms. Age was asked in the post-hoc questionnaire, and not all classroom teachers provided this information.

herself in the third person, he suggested that she may not have a strong self-concept. Mia required assistance with all activities of daily living and required constant monitoring due to severe epilepsy. She was diagnosed with cerebral palsy spasticity.

Ava was 15 years old and showed a general sense of happiness. She took great pleasure in interacting with others through physical or eye contact. However, she persistently avoided touching objects of any kind, making conventional musical instruments inappropriate for her. Her diagnoses included Rett syndrome, apraxia and epilepsy. Her special educational needs included communication in all forms, and according to her classroom teacher, her verbal comprehension was questionable. Her teacher also explained that she was “on the path to intentional communication.”

Alex was 14 years old and joined the music group relatively late after seeing a video of students making music with the instruments we developed. Because of his love for all forms of music, he asked if he could be a part of our group. He had a well-balanced, friendly and cheerful personality. Unlike the other students in our group, he did not have complex disabilities, as he was able to communicate verbally and express his wants and needs through speech. However, due to muscular dystrophy, he required assistance with all activities of daily living. Despite his mobility limitations, he was able to operate his wheelchair independently using a joystick.

4.3 Development of Instruments

Following our design approach (described in section 3), we first identified typical everyday interactions of the children. In the second step, we developed several prototypes of DMIs (see figure 4 for an overview) based on the identified interactions. The instruments were designed to enable collaborative music-making in an ambient style that corresponded to the children’s preferences, which we identified by first listening to different pieces of music with different moods. We discovered that all the children preferred soft and relaxing music. The next step was to test the instruments in individual sessions with the children alone. After that, we used the instruments in collaborative music-making, allowing the children to experiment with different instruments and finally, we collaboratively improvised to a video with the instruments we designed.

Almost all of the DMIs were built in wooden boxes with an embedded Bela computer for sound generation. This made the instruments easy to use by simply plugging in a power supply and a small battery-powered speaker without any additional configuration. In addition, these instruments could be easily stored in school cupboards (which we found to be an important aspect for music teachers [14]).

4.4 Typical Movements and Instrument Design

As typical movements, we identified the categories of moving (touchless), hitting, pressing, shaking, joystick interaction, and shifting. In the following sections, we describe the observations that led to these categories and the instruments associated with them. While all of the DMIs went through several iterations, only the final versions are described here due to space limitations. The iterative process is illustrated by the shaker ball in Figure 5.

Shaking – This category originated from the observation of Liam always carrying a ball-shaped rattle and enjoyed interacting with it. So we started by attaching a ball made of woven wood fibers

to a Gametrak controller and mapping the movement intensity to various loops. The problem was that he didn’t seem interested, which we attributed to a lack of direct haptic feedback like that found in his rattle. So, for the second prototype, we used a similar rattle ball, which he tried to use, but often slipped out of his hand (due to the pull of the Gametrak controller string). As a result, we developed a prototype using an accelerometer and sending data wirelessly to a laptop. Therefore, this particular instrument did not run on Bela due to the wireless connection, which was not natively supported by this system. In future iterations, we are considering integrating a small speaker and a Bela mini directly into the ball. The final prototype featured ocean wave samples that increased in intensity, reflecting Liam’s particular enjoyment of triggering loud noises, while also harmonizing with a calm ambient composition.

Moving (touchless) – This category was created primarily due to Ava’s aversion to touching objects. Our idea was to develop a DMI that works without direct physical contact (like a theremin). The developed instrument used a laser distance sensor to measure the position of a hand or similar object and to map the position to a pentatonic scale. The auditory feedback was a sustained tone (sawtooth with long attack and release and a low-pass filter) that got louder the longer it was held. The selected pitch was visually displayed on an LED strip.

Hitting – This category was based on Leo’s preference for hitting instruments or objects, but Liam and Mia also enjoyed this form of interaction. The instrument consisted of a piezo microphone attached to the wooden surface, which allowed the sound generated by hitting (or stroking, tapping, etc.) to be further processed. For sound shaping, a peak detection algorithm identified the ‘strikes’, each of which triggered a new random pitch. The sound itself consisted of the incoming sound processed through a Karplus-Strong algorithm, creating a metallic, string-like quality. In addition, the instrument included an ultrasonic distance sensor mapped to the cutoff frequency of a low-pass filter to influence the duration of the sound. The LED strip provided visual feedback linked to the amplitude of the sound output and the distance to the ultrasonic sensor.

Pressing – This category was based on several observations. First, we noticed that Ava often used one hand to apply pressure to her other hand. This observation led us to incorporate a pressure sensor into a glove to capture her movement. For auditory feedback, we linked the pressure intensity to ambient loops (pad sounds) in multiple layers. In addition, we implemented a feature where each press of the sensor triggered and modulated a new random pentatonic sound, sounding similar to the theremin. Visual feedback was provided by an LED strip linked to the intensity. The same functionality was used with a pillow. Here we integrated the pressure sensor into a pillow so that students could apply pressure to it to produce sound. Second, many of the children frequently interacted with accessible switches (commonly used in AAC) in their regular classes, so they were familiar with pressing buttons. Thus, we also equipped a box with a 10cm arcade-style button that triggered a short, randomly pitched pentatonic note with each press. To generate the sound, we used the chimes Pure Data patch from Miguel Moreno’s MKMR Pure Data library.² The third instrument

²<https://github.com/MikeMorenoDSP/pd-mkmr>

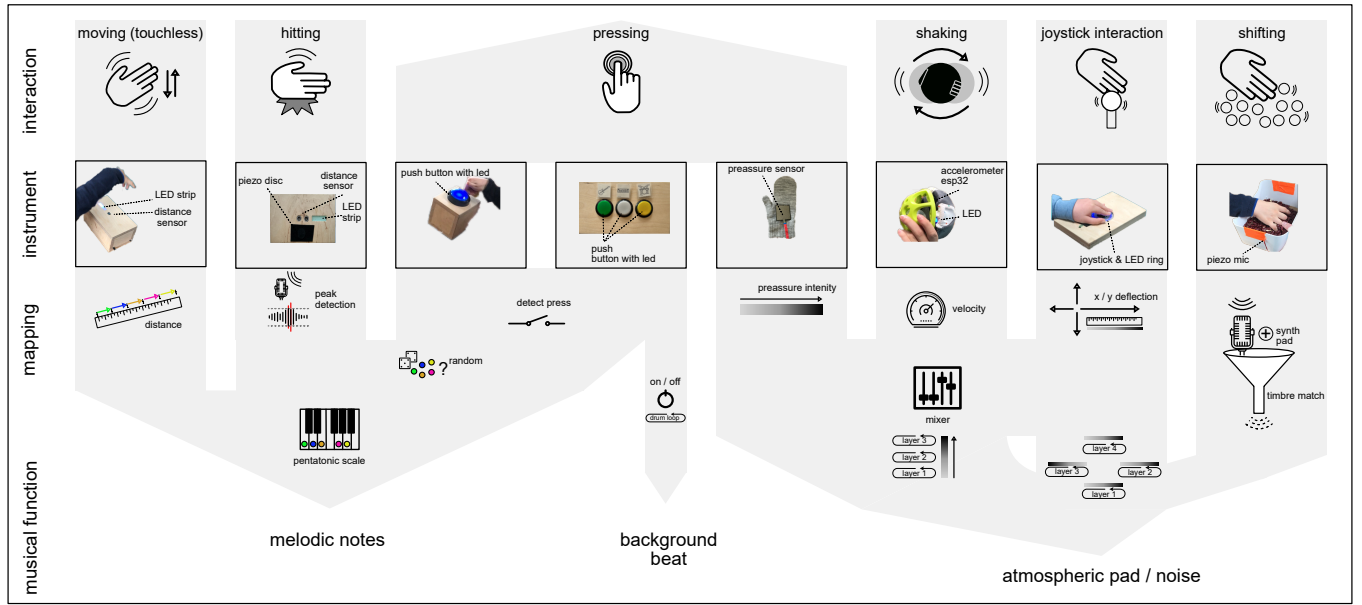


Figure 4: Overview of typical movements and corresponding developed instruments including sound mapping



Figure 5: Different iterations of the shaker-ball

had three buttons and was the most complex one. The first button triggered a deep bass synthesizer with a long envelope and a slowly rising low-pass filter (root note, fourth and fifth steps) that ended after about 20 seconds. The second button produced a higher melody note, and the third button toggled a drum loop on and off.

Joystick Interaction – This category stemmed from Alex’s ability to exert fine motor control over the joystick that he used to operate his wheelchair. Conceptually, we divided the joystick deflection into four quadrants, each controlling the low-pass filter cutoff frequency for individual ambient loops. This allowed for precise control over nuances in the sound and the ability to combine multiple layers. The LED ring around the joystick provided visual feedback on the intensity of the sound and the direction of movement of the joystick.

Shifting – This category was based on two observations. First, Emma’s limited hand-eye coordination, which made it difficult for her to perform precise movements, and her preference for tactile interaction with different materials. Second, the concept of basal stimulation, which includes tactile stimulation through various materials in special education. The instrument is a box filled with beans. The sound generated by movement (shifting) is captured by

a piezo microphone and mixed, similar to a Fourier-based vocoder, with a sound pad composed of low-pass filtered sawtooth oscillators (“timbre match”).

5 EVALUATION

5.1 Observations

When we introduced **Emma** to the instruments (tactile box and pressure pillow), she showed signs of excitement and tension, which her assistant attributed to the unfamiliar situation. After a short time, she adopted a more relaxed posture, first with the button instrument and then with the other instruments. At first, she used the instruments with her assistant guiding her hand, but increasingly she began to show her own impulses. In later group music sessions, she played various instruments, mostly with assistance, but sometimes on her own. Her personal assistant identified the pressure pillow and theremin as being particularly appropriate for her. She described Emma as focused and interested in our music sessions and noted that Emma was very engaged. As a suggestion for improvement, she mentioned that the instruments could use a little more padding.

When trying out the instruments, **Liam** was particularly interested in the shaker ball and the button box. He pressed the button box several times on his own, but showed limited motivation to engage with it for extended periods of time. However, he clearly enjoyed the wireless version of the shaker ball and played it for extended periods during group music sessions. His personal assistant confirmed that the Shaker Ball was “his favorite instrument.” She also described him as quieter than usual during our group music sessions.

Leo showed limited motivation to engage with the instruments for any length of time. His personal assistant mentioned that button interaction worked best for him because he “rejects everything else

and generally tends to throw instruments around instead of making music with them.” He also described him as being a little more restless than usual during our music group because the focus was not on him.

At first, **Mia** operated the instruments only by hand guidance and without much attention. The button did not seem to interest her either. When we offered her the button while singing a song she really liked (Frère Jacques), she happily pressed the button during the part where “Ding, Dang, Dong” was sung and sang along loudly. Afterwards, she continued to play the button with excitement in the group context, frequently singing “Ding, Dang, Dong” as she played. Her personal assistant confirmed that the “Ding, Dang, Dong” button worked exceptionally well for her. She described Mia as more focused and attentive than she was in the regular classroom and noted that Mia participated actively for an extended period of time.

In the one-on-one session with **Ava**, we first experimented with several instruments. However, she did not seem to be interested in any of them. It was not until we tried the pressure glove and the first author of this paper played the theremin with her that we first observed significant levels of entrainment. During this interaction, she showed signs of enjoyment, including laughter, rhythmic movements, and active participation in the music-making process, which persisted for extended periods of time (more than 30 minutes). Her personal assistant also confirmed that the glove was the best instrument for her, although she could not say for certain whether Ava perceived herself as the sound producer during the interaction.

When **Alex** joined the group after watching a video demonstrating the interaction of the pressure sensors, he immediately expressed a strong desire to try it out. He played rhythmically (at first one note per beat, later several in 16th notes) to songs sung by the rest of the group. He was highly motivated to try all the instruments and seemed to enjoy the many different possibilities. His personal assistant indicated in the questionnaire that he particularly enjoyed the joystick instrument because he was able to operate it largely independently without her assistance. She also described him as “more attentive than in regular class.

5.2 Evaluation of the Teachers

Overall, the teachers rated the project as a great success and spoke of significant potential for the students in our project, as well as for other students with similar needs and abilities, as illustrated by the following quote:³

Liam enjoys playing this just as much as his regular rattle, and you can always see it in his face [...]. When he played it for the first time, there was a moment of surprise, he looked to see what was happening here, wow, [...] you could also call it fascination, exactly. So, this has a very high potential for the student, but certainly also for other students who would play the same instrument.

Both teachers particularly emphasized that the customized development and the focus on clear individual forms of interaction (“simplicity”) of the instruments were very appropriate, especially

for the group of students with complex disabilities. The ability to independently and actively trigger sounds was seen as a “great advantage.” In this context, they emphasized the importance of enabling students to experience self-efficacy.

One teacher even described the entire project as a personal “highlight; being able to witness the use of the instruments.” He also described the basic problem that for some students, situations with many simultaneous stimuli can quickly become overwhelming due to their short attention span. However, he noted that this was not the case with the developments from the project because of the harmonious overall result of all the instruments and the design based on the students’ preferences. “But because even in this, in the group as a whole, in the ‘jungle of sounds,’ so to speak, which was fortunately also harmoniously tuned by you, I never had the impression that the severely disabled or perceptually impaired students were overwhelmed.”

The only drawback mentioned by one teacher was that the wooden boxes were still too heavy.

Participation and Exclusion. Teachers described the benefit of the instruments in allowing students to participate in shared music lessons, “because they offer a way for supposedly limited students to participate in this music class or in a creative process, within the group, and that’s what excited me the most about these individual instruments.” In this context, he described regular music classes as often inappropriate for this target group:

The music classes I experienced at our school when I occasionally attended were more aimed at students with better motor skills. They might have built their own sticks or made their own drum and then played it. Of course, all this is not possible for these students, for those with severe multiple disabilities. But with these instruments they can participate.

An important aspect, he said, was that participation in music also had an impact on the perception of other students:

I think what’s good for the fit students with these simple instruments is that they see that the weaker students don’t just have someone to take care of them, so to speak, as a fit student, I take care of my classmate who’s with a caregiver or something, but they see that they’re actively participating in our music ensemble and of course that has a positive effect on the fit students as well. So it’s more about the social component, not just the musical or technical component.

Transferability to Other Groups. The teachers also liked the idea of using the instruments with other students. “Yes, it’s ideal. Of course, other students with similar movement patterns can use it [...] Yes, individuality is there. You made it for this student, but we always have students like this with slightly different variations. So it’s not like when certain students leave, these instruments can’t be used anymore.” Notably, for other students more operational freedom is needed, as “these very simplified controls, like a switch or a sensor, can get boring quite quickly for a fitter student.”

Finally, teachers were asked about additional similar interaction affordances that would be necessary to enable as many students

³The quotes have been translated from German.

with complex disabilities as possible to engage in active music-making. They mentioned eye control, jumping/hopping/waving movements, interaction with the voice, and triggering sounds with body noises such as snapping or clapping.

6 CONCLUSION AND FUTURE WORK

As part of a year-long design project in a SEN school, we developed several DMIs for children with complex disabilities, based on the individual abilities of the children, tested the instruments with the children, and evaluated them in individual settings as well as during group music-making. The design process was based on our proposed idea of a 'Some for Many' approach, using the principle of finding forms of interaction based on the children's typical everyday movements. An underlying assumption was that the musical instruments developed in this way could also provide access to active music-making for other children with similar abilities and needs. Our findings suggest that the design approach is well suited to provide access to active music-making for most of the children in our group and to facilitate diverse aesthetic experiences. This was confirmed by the teachers, who also felt that the instruments would be a good fit for other groups of students with similar abilities and needs. Unfortunately, this assumption could not be further evaluated within the scope of our project and requires further research for validation.

The children's use of the instruments varied greatly. There were children who preferred to use the instrument designed specifically for them (e.g., the shaker ball), while other children responded positively to the different possibilities for interaction and benefited from having a wide variety of musical instruments to choose from. However, in the context of our case study, it also became clear that it is not only the accessibility of musical instruments that is important. As Mia's example shows, the creative use of musical instruments, such as the integration of familiar narratives and meaningful structures, is also important. In addition, Leo's case illustrates that we haven't yet discovered appropriate forms of interaction for all of the children in our case study. It may be one possibility to design DMIs that are appropriate for his preference for "throwaway" interaction. On the other hand, there may be students for whom (collaborative) music-making in general is not desirable. Regardless, our case study has explored an approach that provides access to active music-making for a significant number of students who, according to teacher observations, are typically excluded or passively engaged in conventional (music) education. Therefore, our developments may have considerable potential for promoting inclusion in music education, also because, according to the teachers, making music together has a positive effect on the perception of other students, in line with previous findings [20].

As a qualitative study with a limited sample, it is important to acknowledge that the results are not readily generalizable. Further research on similar projects seems necessary. It is also necessary to explore alternative ways of interaction. Our evaluation has already provided suggestions for possible further development. Basic design criteria for similar projects should include simplicity, direct feedback, clear affordances, ease of use, and out-of-the-box functionality.

Currently, the boxes are still being used by the teacher from our project, which we already consider a great success, as literature has shown that DMIs often remain unused in SEN schools [3, 29]. To facilitate further use, we plan to equip the instruments with an integrated LiPo battery and a built-in speaker, eliminating the need for any hardware setup. We also want to create an optional control unit with a wireless connection to all the boxes, so you can easily configure different tonalities for more flexibility.

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